

Q1 2021 (01 Sep Shift 2)

The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0, x = \frac{\pi}{2}$ is:

(1) $2\sqrt{2}(\sqrt{2} - 1)$

(2) $2(\sqrt{2} + 1)$

(3) $4(\sqrt{2} - 1)$

(4) $2\sqrt{2}(\sqrt{2} + 1)$

Q2 2021 (31 Aug Shift 2)

If the line $y = mx$ bisects the area enclosed by the lines $x = 0, y = 0, x = \frac{3}{2}$ and the curve $y = 1 + 4x - x^2$, then $12m$ is equal to .

Q3 2021 (27 Aug Shift 2)

The area of the region bounded by the parabola $(y - 2)^2 = (x - 1)$, the tangent to it at the point whose ordinate is 3 and the x -axis is :

(1) 9

(2) 10

(3) 4

(4) 6

Q4 2021 (26 Aug Shift 2)

Let a and b respectively be the points of local maximum and local minimum of the function

$f(x) = 2x^3 - 3x^2 - 12x$. If A is the total area of the region bounded by $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$, then $4A$ is equal to .

Q5 2021 (26 Aug Shift 1)

The area of the region

$S = \{(x, y) : 3x^2 \leq 4y \leq 6x + 24\}$ is .

Answer Key

Q1 (1)

Q2 (26)

Q3 (1)

Q4 (114)

Q5 (27)

Q1 (1)

$$A = \int_0^{\pi/2} ((\sin x + \cos x) - |\cos x - \sin x|) dx$$

$$A = \int_0^{\pi/2} ((\sin x + \cos x) - (\cos x - \sin x)) dx$$

$$+ \int_{\pi/4}^{\pi/2} ((\sin x + \cos x) - (\sin x - \cos x)) dx$$

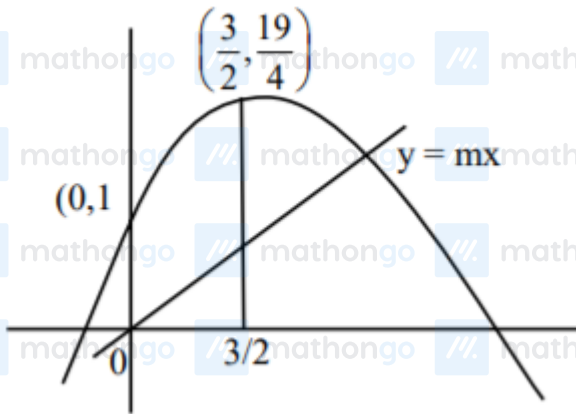
$$A = 2 \int_0^{\pi/2} \sin x dx + 2 \int_{\pi/4}^{\pi/2} \cos x dx$$

$$A = -2 \left(\frac{1}{\sqrt{2}} - 1 \right) + 2 \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$A = 4 - 2\sqrt{2} = 2\sqrt{2}(\sqrt{2} - 1)$$

Option (1)

Q2 (26)



$$\text{Total area} = \int_0^{3/2} (1 + 4x - x^2) dx$$

$$= x + 2x^2 - \frac{x^3}{3} \Big|_0^{3/2} = \frac{39}{8}$$

$$\& \frac{39}{16} = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3}{2} m$$

$$\Rightarrow 3m = \frac{13}{2} \Rightarrow 12m = 26$$

Q3 (1)

$$y = 3 \Rightarrow x = 2$$

Point is (2, 3)

Diff. W.r.t x

$$2(y - 2)y' = 1$$

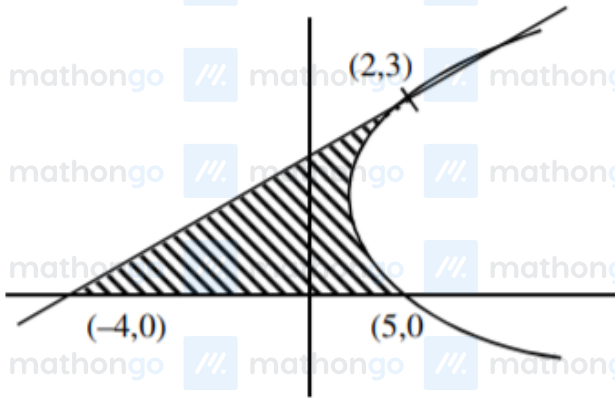
$$\Rightarrow y' = \frac{1}{2(y-2)}$$

$$\Rightarrow y'_{(2,3)} = \frac{1}{2}$$

$$\Rightarrow \frac{y-3}{x-2} = \frac{1}{2} \Rightarrow x - 2y + 4 = 0$$

$$\text{Area} = \int_0^3 ((y-2)^2 + 1 - (2y-4)) dy$$

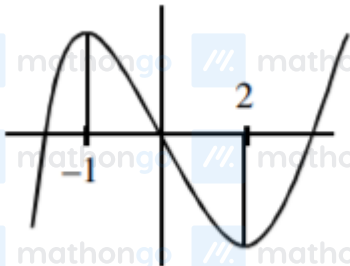
$$= 9 \text{ sq. units}$$



Q4 (114)

$$f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$$

$$\text{Point} = (2, -20) \& (-1, 7)$$

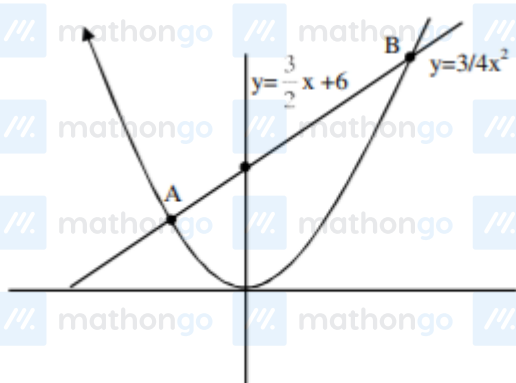


$$A = \int_{-1}^0 (2x^3 - 3x^2 - 12x) dx + \int_0^2 (12x + 3x^2 - 2x^3) dx$$

$$A = \left(\frac{x^4}{2} - x^3 - 6x^2 \right)_{-1}^0 + \left(6x^2 + x^3 - \frac{x^4}{2} \right)_0^2$$

$$4A = 114$$

Q5 (27)



For A & B

$$3x^2 = 6x + 24 \Rightarrow x^2 - 2x - 8 = 0$$

$$\Rightarrow x = -2, 4$$

$$\text{Area} = \int_{-2}^4 \left(\frac{3}{2}x + 6 - \frac{3}{4}x^2 \right) dx$$

$$= \left[\frac{3x^2}{4} + 6x - \frac{x^3}{4} \right]_{-2}^4 = 27$$